

Agile practices adapted to mass-market application development

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ABSTRACT

Mobile app markets have experienced a significant growth during the last year. The increasing number of apps available on markets and the profits that developers and companies obtain are relevant enough to seriously consider the way apps are developed. The always-changing environment in which apps are produced makes agile a convenient methodology to follow. Although agile allows the development team to quickly adjust the requisites to the new customer's needs, there is a lack of research on how agile methodologies can be explicitly adapted to develop mass-market applications. Many references explain how to design and code a mobile app, but there is not much knowledge about other stages in the development process. Adapting an agile methodology to the development of mass-market applications would provide project teams with a clear guide to successfully produce an app without missing any step in the process. This paper describes a case study in which an agile mass-market application development process is followed. The proposed process also considers marketing issues as they are necessary to publicize a mass-market application. This process has been followed and refined for several years in the development of the institutional apps at Carlos III University of Madrid. Copyright © 2014 John Wiley & Sons, Ltd.

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1. INTRODUCTION

Currently, mobile app development is booming to satisfy rapid growth in demand for apps, according to data from Gartner [1]. Although there is a large amount of information about coding apps for mobile devices, there is not much information on how to complete the entire development process.

Agile methodologies are often adopted for mobile software development because of their characteristics. The development of an app should be relatively quick and straightforward. Small teams must deliver new versions in short periods to meet the frequent modifications in customer's needs and expectations [2].

These agile methodologies, however, attempt to strive for universal solutions as opposed to situation appropriate. As a consequence, they usually lack significant aspects that are specific of mass-market applications, such as the co-joint management of a portfolio of apps to develop, mechanisms to involve end users in the process of validating and upgrading the apps, or marketing-related practices to differentiate the app as a product, identify the target customers, and consider other activities to publicize the app and follow up the marketing efforts, among others.

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To tackle these problems, a software development process based on agile methodologies was specifically tailored to develop mass-market applications. We decided to use agile because it adapts quickly to changes, which is very important in the highly dynamic and competitive market of mobile apps. The inclusion of marketing activities in the process is considered innovative because agile methodologies do not usually include marketing phases in their processes.

This paper presents the approach adopted to define and evolve a method to develop mass-market applications (specifically mobile apps) combining agile development with marketing-related practices. The paper also describes the processes that compose the method with their corresponding activities and the appropriate roles for each one of these activities. Finally, several relevant lessons learned from the implementation of the proposed method are discussed.

2. BACKGROUND

Agile methodologies have been used since the 1990s for software development. Because of the growth of light applications oriented to mobile devices, agile is currently one of the most adopted methodologies for mobile software development [3]. The characteristics of all agile methodologies are based on the Agile Manifesto [4], which emphasizes individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation, and responding to change over following a plan.

In the last years, numerous agile methodologies have appeared with different characteristics each [5]. However, all of them share the following attributes: incremental, cooperative, straightforward, and adaptive [6]. Incremental refers to small software releases, with rapid development cycles. Cooperative refers to a close customer and developer interaction. Straightforward implies that the method itself is easy to learn and to modify and that it is sufficiently documented. Finally, adaptive refers to the ability to make and react to last-moment changes.

These characteristics make agile methodologies suitable for mobile app development [2]. The development of an app should be relatively quick and straightforward. Small teams must deliver new versions in short periods to meet the frequent modifications in customer's needs and expectations. And this must be performed maintaining an appropriate product quality and reducing development costs.

Some popular agile methodologies are eXtreme Programming [7], Scrum [8], Kanban [9], and a hybrid approach called Scrumban [10]. Table I summarizes the most relevant practices proposed by these agile methodologies.

Despite the fact that XP, Scrum, Kanban, and Scrumban are often followed for the development of mobile apps, these agile methodologies do not consider several aspects important for the development of mobile apps oriented to mass markets.

The most important aspect that agile methodologies lack is marketing, essential for a mass-market application to be more likely to succeed [11]. As it was mentioned earlier, mass-market application development brings a change in the philosophy of traditional software development. A mass-market application does not usually provide fixed profits from its sale to a particular client, but it depends on its price and the number of downloads (i.e., buyers).

Marketing must not be a single process completely isolated from development. It should be integrated throughout the project in its different phases, from the conception of the app to its launch in the market of mobile apps [12].

Firstly, when an idea is expressed in the form of a description of the app, several points should be considered, such as the current trends in the market or the differential value that makes the app distinguishable from other apps on the market. It is also important to identify the audience the app is aimed at, because the outcome will depend on the number of customers who find the app useful. Furthermore, it is necessary to identify the competitors of the app to emphasize those values that make our app different from theirs and also to learn from them.

Before deciding to develop the app or not, a market analysis is also needed to obtain insights about the viability of publishing our application in the market of mobile apps. So, it is necessary to know the market size and its growth rate, assess its profitability, and compare the expected costs and profits to

Table I. Agile software development practices.

	Product definition and requirements management	Project and iteration planning	Development	Quality assurance	Project control and oversight
eXtreme programming	Story cards, user stories	Adaptive planning, task list, frequent releases	Metaphors, simple design, refactoring, CRC cards, pair programming, continuous integration, coding standards	Test-driven development, customer tests	Sustainable pace, short releases
Scrum	Product backlog	Timeboxed sprints, frequent releases, sprint backlog, self-organized and self-managed teams, story points, planning poker, sprint planning meetings			Big visible charts, burn up-down chart, daily stand-up, sprint review, iteration retrospective, definition of done
Kanban	User story cards	Kanban board (visualize workflow), limit work in process (WIP)			Measure and improve flow, look for bottlenecks
Scrumban	User story cards	Simplified Kanban board			Same than Kanban, but prescribes Scrum daily meeting, sprint review and retrospective

estimate the return on investment (ROI). The identification of these aspects is important to obtain relevant information needed for any decision-making situation in the project.

While the app is being developed, a marketing plan must be created to later publicize the app and make it well-known to the public. The plan must set the goals of the marketing campaign, evaluate the strengths, weaknesses, opportunities, and threats involved in the project, establish the channels through which the app will be publicized, and estimate the expected results of the marketing effort and the appropriate indicators to measure those results [13].

Before submitting the app to the market, some marketing-related information must be prepared as it will be needed to publish the app. Elements such as the screenshots of the app, a keyword list or the description of the app are key to sell the app because many customers will decide whether to buy the app or not depending on this information.

Finally, once the app is already published in the market, feedback that users provide about the developed app, besides other relevant data, must be gathered to analyze if the app is meeting the goals previously set in the marketing plan or not. It is important to assess the advertising effectiveness [13] to obtain insight into how the marketing campaign is performing, not only for evaluation purposes but also to take a proactive approach to all our account management.

To conclude this section, we have presented some aspects that agile methodologies lack when used for developing mass-market applications. To fill this gap, a process based on agile methodologies was specifically adapted to the development of this kind of applications. The next sections describe the context of a case study in which this adapted process is being followed, a brief description of the method and the lessons learned so far.

3. DEFINING MISEL SOFTWARE DEVELOPMENT PROCESS

miSEL Software Development Process (miSEL-sdp) was defined in the scope of a case study that consisted of the implementation, coaching, analysis, and evaluation of the agile and marketing-related practices to develop mobile apps by miSEL group at Carlos III University of Madrid.

miSEL aims at the research and development of mobile apps to promote the quality of life in a university context to support different activities related to daily life, teaching/studying, or navigation in campuses and large buildings. Moreover, miSEL provides development workforce to implement mobile solutions related to several research initiatives of the Institute for Automobiles Security at Carlos III University of Madrid.

A representative list of mobile apps developed by miSEL developed during the definition of miSEL-sdp is presented in Table II. These apps have been developed and upgraded between 2011 and 2014 and can be downloaded from App Store or Google Play.

miSEL group is composed of a team leader having 10 years of experience in the management of agile software projects and four junior software engineers with 2–4 years of experience in mobile software development, including technologies such as iOS, Android, Java, Web services, SQL databases, HTML5, JQuery, and PHP among the most relevant implemented in the previously mentioned apps. The responsibility of this group of participants was the application of specification, design, and development practices recommended by the researchers guiding this research work. Moreover, miSEL collaborated continuously with five marketing and communication experts in charge of supporting miSEL team members in the implementation of marketing-related activities.

In parallel with the development of the apps, the group of researchers (composed of three people) contributed to the case study by recommending, monitoring, and analyzing how the proposed practices were being applied. The researchers have more than 7 years of experience teaching software project management and agile software development.

At the beginning of the case study, several practices related to product identification and definition, requirements management, mobile app testing, iteration control and tracking, and project management were recommended to miSEL software development team. Periodically, the miSEL team members indicated whether they had been able to implement particular practices, how they had implemented them, the difficulties involved, and their opinions on which practices contributed the most to the effective development and management of the considered apps.

Table II. Apps developed during the case study.

App name	Platforms	App type	Purpose
UC3M	iOS/Android	University life	Provision of general daily information of the university, such as directory, studies, news, and videos
i-3 Magazine (iUC3M)	iOS	Magazine	Monthly publication of the institutional magazine by Carlos III University
UC3M Agenda	iOS/Android	University Life	Information of the institutional university agenda connected with the personal agenda
UC3M Mapas	iOS/Android	Indoor location	Indoor navigation solution to find a classroom or office
UC3M Students	iOS/Android	University life	Provision of relevant information related to the daily life of students at Carlos III University
UC3M Emergencies	iOS/Android	University life	Provision of recommendations in case of different type of emergency situations at the university
TECHFEST-UC3M	iOS/Android	Conference management	Provision of agenda, indoor navigation solution, videos, and social interaction in UC3M TechFest conferences
ITGSM 12	iOS/Android	Conference management	Provision of agenda, indoor navigation solution, videos, and social interaction in ITGSM 2012 conference
mTeachPlan ^a	iOS	Teaching and learning support	Planning the teaching sessions in a course to enable the interaction with students through tweets, questionnaires, videos, and presentations
mTeachClass ^a	iOS/Android	Teaching and learning support	Enabling the interaction with students during a lesson through tweets, questionnaires, videos, and presentations
mClassTest ^a	iOS/Android	Teaching and learning support	Feedback from students to teacher in terms of questionnaire answers and/or tweets
mTeachEval ^a	iOS/Android	Teaching and learning support	Collaborative evaluation of students assignments
fBrake ^b	Android	Compulsory vehicle inspection	Solutions to perform the brake test of trucks under European regulations
Traffic for Kids	iOS	Education in traffic regulations	simulating full load conditions required during compulsory vehicle inspections
Traffic for Teens	iOS	Education in traffic regulations	Provision of training in traffic regulations specifically focused on teens
Car Assistance	iOS/Android	Automobiles security	Provision of training in traffic regulations specifically focused on kids
Technical Inspection of Vehicles	Android	Compulsory vehicle inspection	Giving recommendations about what to do in case of a car accident
			End-user support to prepare car compulsory vehicle inspections

^aAvailable only at UC3M Internal Apps Market.^bAvailable only at fBrake portal for CVI stations in Spain.

At the beginning of 2012, several practices related to marketing purposes were included in miSEL-sdp method. These practices were related to market analysis, marketing plans, and marketing-related feedback. Several adaptations were introduced in the method in order to incorporate effectively new required tasks.

At the end of 2012 and 2013, retrospective meetings were carried out. Researchers facilitated this kind of meeting by asking miSEL team members to discuss on the benefits and difficulties to implement the proposed practices. The specific questions addressed in these meetings were the most useful practices, the activities they did not want to include in new projects and activities that could be improved, identifying an appropriate way to improve them. After a brainstorm to set an initial list of ideas, team members voted on specific items to focus on during the next period.

Evidences gathered to support the analysis of the implementation of the miSEL-sdp practices included the registry of the project activities (provided by the project management tool) and the notes obtained from the retrospectives and informal interviews with miSEL team members.

Once the data gathering phase finished, the last analysis of miSEL-sdp was carried out in order to provide the current definition of the development process.

4. MISEL SOFTWARE DEVELOPMENT PROCESS

This section briefly describes a process based on agile methodologies specifically adapted to the development of mass-market applications named miSEL-sdp. A detailed description of the different processes can be found in [14]. Figure 1 shows the flow through the different processes considered in the proposal and the work products involved in those processes.

Apps Portfolio Management aims at selecting the app to develop. To do so, the project team has to identify the customer's needs or the opportunities that arise to develop an app. A portfolio of apps that may satisfy those needs or take advantage of those opportunities is established and prioritized; the highest-priority app on the list will be the one selected for its development.

The *Conceptual Definition of the App* states its features and restrictions; the strengths of the app should represent a differential value for the target audience when compared with competitors. These

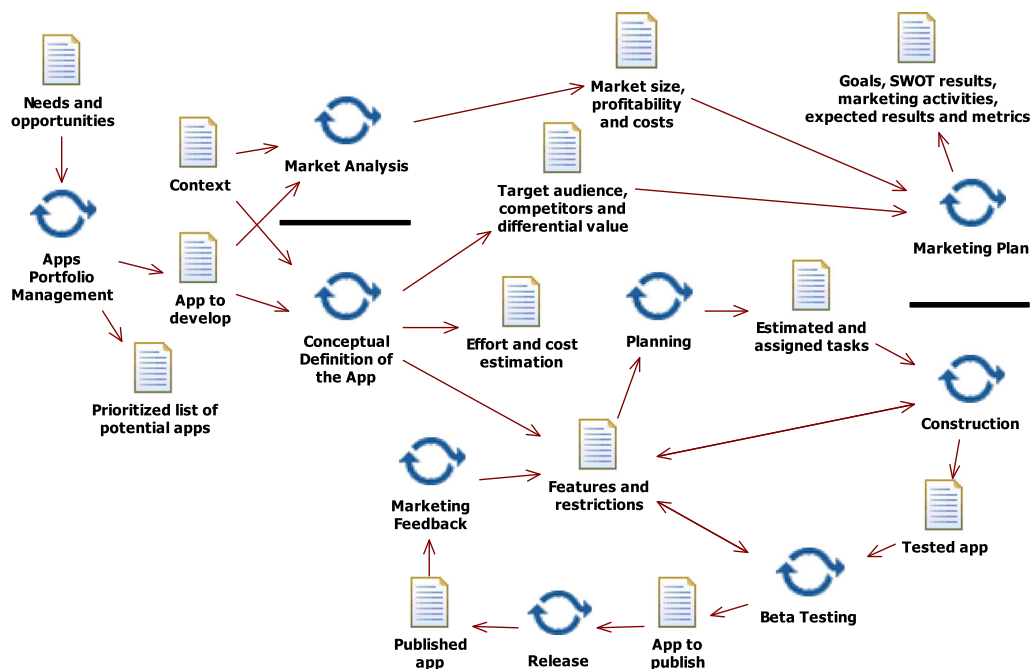


Figure 1. Processes adapted to the development of mass-market applications.

characteristics and a first estimation of the effort and cost for the project will help decide whether the development of the selected app is feasible or not.

The *Market Analysis* is carried out in parallel with the Conceptual Definition of the App to determine whether the market is sufficiently attractive to launch the app. To estimate the ROI, the size of the market and its growth rate must be measured, and potential buyers must be identified as well as how much they will be willing to pay for the app, how much it will cost to produce the app, and why they will want to buy our app and not others.

Planning should be based on the features and restrictions previously identified, which will help create a list of tasks to be carried out by the different roles involved in the project to obtain a successful result. Every task has to be estimated in terms of time and cost and compared with the estimation obtained in the Conceptual Definition of the App process for the whole project; some tasks might be re-estimated in consequence.

Once the Planning is carried out, *Construction* of the app starts. This iterative process not only covers coding the app but also includes its design and forthcoming tests to validate the requirements. When feedback received from the execution of the tests is positive, this process finishes and a beta version of the app is distributed to the stakeholders; otherwise, the cycle starts again with a re-design to fix the issues found in the tests.

The *Marketing Plan* aims at specifying a plan to publicize the app, which should consider several aspects. First, an analysis of the strengths, weaknesses, opportunities and threats (SWOT analysis) will provide relevant information of the app when compared with similar ones. The marketing plan should establish the communication activities needed to promote the app through different channels, such as a website and social networks. Expected results of the marketing effort and metrics to monitor the marketing campaign must be also determined.

The objective of the *Beta Testing* is that the tested app (beta version) is validated by the stakeholders to find defects missed in the tests executed by the development team during the construction process. If the stakeholders provide satisfactory results, the app is ready to be released on the market of mobile apps; otherwise, the list of features to be implemented is updated.

The aim of the *Release* process is to have the app available in the market for customers to buy it and thus obtain the expected profits. Markets of mobile apps require a packaged app and some information about the app. In addition, markets usually perform a validation process to assure that the app meets the criteria established for publishing apps in that market. If the app does not pass the market validation, the project team receives feedback from the market reviewers regarding the issues found that made the app being rejected.

The *Marketing Feedback* gathers the comments that users provide about their experience with the published app, besides other relevant data, to analyze them. The analysis of these data will be useful to identify issues to fix in the current version of the app, to define new functionalities that may be included in future versions, and to track marketing activities. As a consequence, that information should go back to the Planning process.

5. LESSONS LEARNED

The lessons learned during the definition of miSEL-sdp fall into the following categories.

5.1. Multiple disciplines and practices to define a new product

During the considered case study, in order to increase the attractiveness of the mobile apps and their capacity to obtain the expected number of downloads and continuous use, several activities were needed to define the mass-market application:

- identification of target audience;
- determination of information consumption patterns;
- identification of most used devices (smartphones, tablets, etc.) and the trends in user interaction preferences and patterns; and
- analysis of competitors.

The correct implementation of these activities requires the effective coordination of multidisciplinary teams. In the case study considered, this situation led to larger periods to define a product because of issues related to agenda coordination and availability of high-specialized roles related to marketing of mobile apps. In this sector, the timespan between the conception of the idea and the product available in the market is a critical success factor, preventing a wider dissemination of apps such as UC3M Maps and other commercial developments from this app.

During the first attempts to involve participation of marketing and communication experts, the authors identified a clear tendency to a literal translation of preexisting web pages to a mobile app as in the case of UC3M App. After launching several upgrades, these experts realized that most downloaded and used apps usually provide very specific information, adapted to the specific device and user context, in few seconds and taking few steps. The introduction of this philosophy required some re-work in this app, but it enhanced rapid development and effective distribution through the key audience as in the case of UC3M Agenda, UC3M Students, TechFest, and ITGSM 12 conference apps.

Finally, developers and marketing and communication experts agreed to consider the effective analysis of competitors as a critical success factor to identify opportunities, competitive advantages, and differentiation from other apps. Even more, it was considered very useful for this purpose to analyze indirect competitors, that is, alternative apps that satisfy the same need than our app but for a different target audience (e.g., mTeachPlan, mTeachClass, mClassTest, and mTeachEval cases).

5.2. *Size and effort estimation of mobile apps*

Similarly to other methods specifically focused in the development of mobile apps [15], miSEL-sdp provides guidance for requirements engineering and interaction design for mobile apps and also offers interfaces to other activities, such as visual design, architectural design, implementation, and testing. Nevertheless, a problem partially addressed is the lack of widespread methods to estimate the effort and cost required to develop a mobile app.

The requirement specification of mobile apps is usually based on the creation, refinement, and evolution of mock-ups and prototypes. In these conditions, miSEL team members experimented that it is difficult to use traditional software estimation methods to determine mobile apps size and effort. In order to address this issue, widespread estimation methods such as planning poker and story points [16] were introduced.

Currently, agile estimation methods are based on similarity because effort is estimated by assessing the effort of one task relative to that of another. In accordance to [17], the information from retrospectives and informal interviews to miSEL members indicated that relative estimation could result in biased assessments of similarity and overly optimistic effort estimates.

In the same way than discussed in [18], several specific needs, not addressed in this research work, were identified: (i) facilitate effort estimation from the information included in the mock-ups guiding the development efforts; (ii) create a common understanding of the story points concept; (iii) provide effective ways to classify and categorize the historical data on size, effort, and cost; (iv) improvement of benchmarking mechanisms and data; and (v) validate empirically estimation models.

5.3. *New challenges for solution design*

The remarkable growth of mobile apps has spawned a number of different profit models, none of which has yet proven to be the gold standard for most developers. Paid apps have worked for a select few, while in-app purchases and ad-supported models are gaining popularity.

To implement effectively these monetization strategies, miSEL members found it essential to use statistics or explicit user queries to identify user's locational and activity habits in order to provide useful information for ad matching. In the same way indicated by [19], relevant issues to discuss during app design include the following: (i) logs of user behavior; (ii) explicitly provided by the user through queries posed to him over time; (iii) periodic generation of a distilled summary of the user's habits and intentions when using the app; and (iv) queries to the user profile on demand to select the most appropriate ad to a use.

During the case study, the experience obtained in several apps (UC3M Students, UC3M Maps, UC3M Agenda, and i-3 Magazine) indicates that the early identification of the information to consider for user profiling is important to design properly the app architecture. In UC3M Agenda,

important re-work had to be carried out to include this kind of services because it was decided in the mid of the development phase, while the other apps could include these mechanisms without re-work because of the early identification of user profile requirements.

The implementation of effective feedback functionality based on user habits when using the app also provides critical and relevant information to communication and marketing managers to determine the right information to distribute using specific apps.

5.4. Continuous improvement

During the case study considered in this research work, the implementation of retrospectives and mechanisms to enhance the evaluation and improvement of miSEL-sdp method were essential to increase the team performance.

The foundations of miSEL-sdp method come from the values and principles of the Agile Manifesto. Unfortunately, there are no official criteria for determining whether or not a project team is taking an agile approach, so teams may claim to be agile when they are not, and vice versa.

To achieve this issue and facilitate the miSEL team continuous improvement, several evaluation areas from [20] were adopted: (i) mechanisms and capacity to regularly produce value for stakeholders; (ii) verification and validation mechanisms assuring the product quality; (iii) mechanisms enabling active involvement of stakeholders; (iv) team self-organization capacity; and (v) effective mechanisms to plan and control iterations and projects.

6. CONCLUSIONS

This paper shows how agile methodologies can be specifically tailored to develop mass-market applications. Although agile methodologies adapt quickly to changes, which is very important in the highly dynamic and competitive market of mobile apps, they lack significant aspects important for the development of mobile apps oriented to mass markets.

The software development process proposed in this paper includes the co-joint management of a portfolio of potential apps to develop, mechanisms to involve end users in the process of validating and upgrading apps, and marketing-related practices to differentiate the app as a product, identify the target customers and possible competitors, publicize the app through different channels to make it well-known to public, and follow up the marketing efforts, among others.

The proposed process has been followed and refined by the miSEL group at Carlos III University of Madrid (UC3M) for several years. The use of this process has led to the successful development of several mass-market applications for Android and iOS, including some of the official apps launched by UC3M.

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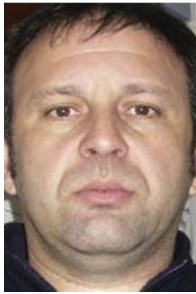
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